

Ziheng Cheng

✉ ziheng_cheng@berkeley.edu 🏠 Homepage 🌐 github.com/Alexczh1

EDUCATION

Ph.D. in Industrial Engineering and Operations Research

Sep, 2024-

UC Berkeley, CA, USA

B.S. in Mathematics

Sep, 2020-Jun, 2024

Peking University, Beijing, China

RESEARCH INTEREST

Generative AI, Optimization, Machine Learning Theory, Game Theory, Statistics

PUBLICATIONS & MANUSCRIPTS

(* stands for equal contribution)

- OVERT: A Benchmark for Over-Refusal Evaluation on Text-to-Image Models
Ziheng Cheng*, Yixiao Huang*, Hui Xu, Somayeh Sojoudi, Xuandong Zhao, Dawn Song, Song Mei
(preprint, under review)
- Provable Sample-Efficient Transfer Learning Conditional Diffusion Models via Representation Learning
Ziheng Cheng, Tianyu Xie, Shiyue Zhang, Cheng Zhang
(preprint, under review)
- Data-Efficient Training by Evolved Sampling
Ziheng Cheng, Zhong Li, Jiang Bian
(preprint, under review)
- Semi-Implicit Functional Gradient Flow for Efficient Sampling
Shiyue Zhang*, **Ziheng Cheng***, Cheng Zhang
(preprint, under review)
- Convergence of Distributed Adaptive Optimization with Local Updates
Ziheng Cheng, Margalit Glasgow
(ICLR 2025)
- Functional Gradient Flows for Constrained Sampling
Shiyue Zhang*, Longlin Yu*, **Ziheng Cheng***, Cheng Zhang
(NeurIPS 2024)
- Kernel Semi-Implicit Variational Inference
Ziheng Cheng*, Longlin Yu*, Tianyu Xie, Shiyue Zhang, Cheng Zhang
(ICML 2024)
- Reflected Flow Matching
Tianyu Xie*, Yu Zhu*, Longlin Yu*, Tong Yang, **Ziheng Cheng**, Shiyue Zhang, Xiangyu Zhang, Cheng Zhang
(ICML 2024)
- The Limits and Potentials of Local SGD for Distributed Heterogeneous Learning with Intermittent Communication
Kumar Kshitij Patel, Margalit Glasgow, Ali Zindari, Lingxiao Wang, Sebastian U Stich, **Ziheng Cheng**, Nirmal Joshi, Nathan Srebro
(COLT 2024)
- Momentum Benefits Non-IID Federated Learning Simply and Provably
Ziheng Cheng*, Xinmeng Huang*, Pengfei Wu, Kun Yuan
(ICLR 2024)
- Particle-based Variational Inference with Generalized Wasserstein Gradient Flow
Ziheng Cheng*, Shiyue Zhang*, Longlin Yu, Cheng Zhang
(NeurIPS 2023)
- Joint Graph Learning and Model Fitting in Laplacian Regularized Stratified Models
Ziheng Cheng*, Junzi Zhang*, Akshay Agrawal, Stephen Boyd
(preprint, under review)

RESEARCH EXPERIENCE

Reinforcement Learning & Stochastic Control

May, 2025 -

Advisor: Prof. Xin Guo, UC Berkeley

- Studied reinforcement learning in continuous time with deterministic policy.
- Established a mathematical rigorous framework for SDE-driven reinforcement learning for the first time.
- Achieved the state-of-the-art performance in various benchmarks.

Safety Alignment in Generative AI

Oct, 2024 - May, 2025

Advisor: Prof. Song Mei, UC Berkeley

- Studied the safety landscape in generative AI, especially on text-to-image models.
- Constructed an automatic pipeline based on LLM to generate large scale evaluation benchmarks.
- Analyzed over-refusal phenomenon and safety-utility tradeoff in T2I models for the first time.

Distributed Optimization & Federated Learning Theory

Mar, 2023 - Oct, 2023

Advisor: Prof. Tengyu Ma, Stanford University; Prof. Kun Yuan, Peking University

- Studied the benefits of local iterations to reduce communication in distributed setting.
- Proposed a distributed adaptive optimization algorithm based on gradient-clipping and Adam.
- Achieved the convergence result of Adam in distributed setting for the first time.
- Proved that momentum can accelerate the convergence of classic FL algorithms without additional assumption.

Diffusion Model & Learning Theory & Sampling Algorithm

May, 2022 - Oct, 2024

Advisor: Prof. Cheng Zhang, Peking University

- Studied sample complexity theories of transfer learning diffusion models.
- Studied particle-based variational inference methods to accelerate MCMC sampling.

INDUSTRY EXPERIENCE

Microsoft Research Asia

Oct, 2023 - May, 2024

Research Intern

Manager: Zhong Li, Jiang Bian

Data Selection for Efficient Training LLM

AWARDS AND HONORS

Awards

- Honorable Mention in Alibaba Global Mathematics Competition 2022, 2023, 2024
- Bronze Medal in S.-T. Yau College Student Mathematics Contest 2022
- Meritorious Winner in Mathematical Contest in Modeling 2022

Honors

- May-Fourth Scholarship (top scholarship in Peking University, 0.5%) 2023
- National Scholarship (top 0.2% nation-wide) 2021
- Merit Student of Peking University 2021-2023